

Detecting Timeout Effects in the NBA via the Generalized Likelihood Ratio Test

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Abstract

Does an NBA timeout actually do anything to a game? We pose that question as a series of one-sample Gaussian signal-detection problems, one per impact metric, timeout type, and game-state slice. For three impact metrics derived from the cdnba 2020–2025 play-by-play feed – excess net points, excess points-per-possession (PPP), and win-probability added – we test $H_0 : \mu = 0$ against $H_1 : \mu \neq 0$ using the generalized likelihood ratio statistic $G = -2 \log \Lambda$, which under H_0 is asymptotically χ_1^2 . The tests are run separately for coach full timeouts, coach challenges, and inferred mandatory television timeouts, and within each subtype we slice by score margin, streak, game phase, home court, and regular season vs. playoffs. The point-differential metric is null in the aggregate but surfaces a clean regression-to-mean pattern once we stratify by margin or streak. The win-probability metric, in contrast, rejects H_0 by enormous margins: full coach timeouts cost the calling team about 3.9 percentage points of WP across a ± 6 possession window, challenges add about 1.6 pp, and television timeouts cost 1.3 pp – broadly aligned with the prior causal literature. Of 163 stratified GLRTs, 69 reject at $\alpha = 0.05$ (42.3%), about $8\times$ what we would expect if the null held everywhere.

Keywords: Generalized likelihood ratio test, signal detection, sports analytics, NBA, timeouts, win probability

1 Problem Statement

For every coach-called timeout, the NBA play-by-play feed records the state of the game on either side of the stoppage. Whether anything in those few possessions actually traces back to the timeout itself is genuinely contested (Assis et al., 2020; Gibbs et al., 2021; Weimer et al., 2023; Qiu et al., 2025). Rather than try to estimate a causal effect directly, we ask a narrower question: for each of three per-event impact metrics, taken within a fixed slice of game state, is the population mean zero?

Let $X_1, \dots, X_n$ be the impact-metric values across timeouts within some stratum – say, full coach timeouts called while leading by 6–15 points – and assume X_i iid $\mathcal{N}(\mu, \sigma^2)$ with both parameters unknown. We test

$$H_0 : \mu = 0 \quad \text{vs.} \quad H_1 : \mu \neq 0. \quad (1)$$

A rejection says the timeout type, conditioned on the stratum, is associated with a non-zero shift in the metric.

2 Data and Metrics

Our data is the cdnba enriched play-by-play feed for the 2020–2025 seasons (regular season and playoffs): 4,183,347 rows across 7,457 games and 1,480,697 possessions. The feed does not label mandatory TV timeouts, so we inject them with a deterministic rulebook procedure and validate the injection against the explicitly labeled 2013–2016 nbastatsv3 feed, where it scores F1 between 0.88 and 0.92. After injection the corpus contains 74,236 `full` coach timeouts, 1,396 `challenge` events, and 14,538 inferred `official_inferred` (TV) timeouts.

For each timeout event we look at a window of $W = 6$ possessions before and W after, and compute three metrics from the calling team’s perspective.

- Q1. `excess_net_post`** Calling-team net points (for – against) in the post window, minus the season-PPP-gap expectation $(\text{PPP}_{\text{calling}} - \text{PPP}_{\text{opponent}}) \cdot W/2$. $\mu = 0$ corresponds to the calling team performing at the season expectation.
- Q2. `excess_ppp_for_post`** Calling-team PPP in the post window minus that team’s full-season PPP. $\mu = 0$ means the team scores at season pace.
- Q3. `wp_added_calling`** Change in calling-team win probability across the $\pm W$ possession window, where $\text{Pr}(\text{home wins})$ is the output of a logistic-regression model trained on possession-start snapshots with the ten features score margin, period, seconds remaining, clutch indicator, possessing-team-is-home indicator, streak, playoff indicator, home win-percentage, away win-percentage, and the home-minus-away win-percentage difference. $\mu = 0$ means the window had no net WP movement for the calling team.

The WP model achieves AUC 0.844 and Brier score 0.161 on a held-out-by-game test set ($n = 59,860$).

For each metric we run the GLRT separately for the three timeout subtypes and stratify on five context columns from the project README: `is_home_calling`, `IsPlayoff`, `time_bucket` (Q1, Q2, Q3, Q4_early, Q4_clutch, OT), `margin_bucket` (seven buckets from down-15 to up-15 from the calling team’s perspective), and `streak_bucket` (seven buckets from a 10-pt opposing run to a 10-pt own run).

3 Methods: GLRT Derivation

For $X_1, \dots, X_n$ iid $\mathcal{N}(\mu, \sigma^2)$ the log-likelihood is

$$\log L(\mu, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2.$$

Under H_0 ($\mu = 0$) the MLE is $\hat{\sigma}_0^2 = \frac{1}{n} \sum x_i^2$; under H_1 the MLEs are $\hat{\mu}_1 = \bar{x}$ and $\hat{\sigma}_1^2 = \frac{1}{n} \sum (x_i - \bar{x})^2$. The supremum log-likelihood under each hypothesis evaluates to $-\frac{n}{2}(1 + \log(2\pi\hat{\sigma}_*^2))$, so the generalized likelihood ratio is

$$\Lambda = \frac{\sup_{\sigma^2} L(0, \sigma^2)}{\sup_{\mu, \sigma^2} L(\mu, \sigma^2)} = \left(\frac{\hat{\sigma}_1^2}{\hat{\sigma}_0^2} \right)^{n/2}. \quad (2)$$

Metric	Subtype	n	$\bar{x}$	G	p	sig.
Q1: excess net post	full	74,236	-0.011	0.98	0.32	ns
	challenge	1,396	-0.112	2.16	0.14	ns
	official_inferred	14,538	-0.003	0.01	0.90	ns
Q2: excess PPP for post	full	74,236	+0.007	7.12	0.008	**
	challenge	1,396	-0.021	1.42	0.23	ns
	official_inferred	14,538	+0.013	5.59	0.018	*
Q3: WP added (calling)	full	74,236	-0.039	10,881	$< 10^{-300}$	***
	challenge	1,396	+0.016	34.3	4.7×10^{-9}	***
	official_inferred	14,538	-0.013	245.2	$< 10^{-50}$	***

Table 1: Aggregated GLRT results by metric and timeout subtype. Significance: ns ($p \geq 0.05$), * ($p < 0.05$), ** ($p < 0.01$), *** ($p < 0.001$).

Using the identity $\hat{\sigma}_0^2 = \hat{\sigma}_1^2 + \bar{x}^2$ we obtain the test statistic

$$G = -2 \log \Lambda = n \log \left(1 + \frac{\bar{x}^2}{\hat{\sigma}_1^2} \right), \quad (3)$$

which by Wilks' theorem is asymptotically χ_1^2 under H_0 . We reject H_0 at level $\alpha = 0.05$ when $G > \chi_{1,0.95}^2 \approx 3.841$. Equivalently, with $s^2 = \frac{n}{n-1} \hat{\sigma}_1^2$ and the one-sample t -statistic $t = \bar{x}/(s/\sqrt{n})$,

$$G = n \log \left(1 + \frac{t^2}{n-1} \right) \xrightarrow{n \rightarrow \infty} t^2,$$

so the GLRT and the standard t -test are asymptotically equivalent; we report both throughout for cross-checking.

4 Results

4.1 Aggregated GLRT Results

Table 1 reports the GLRT statistic for each (metric, subtype) pair pooled over the entire cdnba era.

Q1. Pooled across the league, Q1 is dead null: $G = 0.98$, $p = 0.32$ for the most populous `full` subtype. This matches the null result [Assis et al. \(2020\)](#) report and is consistent with plain regression to the mean. Timeouts are mostly called from a deficit, so the gap between pre-timeout net ($\bar{x}_{\text{pre}} = -1.73$ pts) and post-timeout net ($\bar{x}_{\text{post}} = -0.02$ pts) gets absorbed by the season baseline once we subtract the PPP gap.

Q2. Stripping out the opponent's contribution helps. Q2 rejects H_0 for the `full` subtype ($+0.007$ PPP, $p = 0.008$) and for the `official_inferred` subtype ($+0.013$ PPP, $p = 0.018$). The effects are small but real: roughly 0.7% of a possession's worth of scoring above season pace for coach timeouts, and 1.3% for TV timeouts.

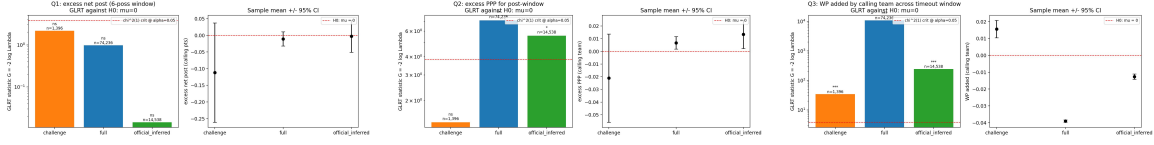


Figure 1: Aggregated GLRT statistics and means for Q1 (left), Q2 (middle), and Q3 (right). The dashed red line on the left panel of each plot is $\chi^2_{1,0.95} = 3.841$.

Q3. Q3 rejects H_0 for every subtype, by huge margins. Full coach timeouts cost the calling team 3.9 WP percentage points on average across the ± 6 possession window ($G = 1.09 \times 10^4$); challenges *add* 1.6 pp ($G = 34$, $p < 10^{-8}$); TV timeouts cost 1.3 pp ($G = 245$). The challenge sign is unsurprising – challenges are filed when a team thinks a call went against them, so the post-window typically reflects a restored possession or extra free throws. The negative sign on full coach timeouts is the more striking finding, and lines up with the slightly negative ATT [Gibbs et al. \(2021\)](#) reported via genetic matching.

Figure 1 plots the GLRT statistics on a log scale next to the sample mean and a 95% confidence interval. The orders-of-magnitude gap between Q1 and Q3 is most of the story here: measuring impact in WP units rather than raw point swings amplifies the statistical signal by several orders of magnitude.

4.2 Stratified by Pre-Timeout Streak

The streak bucket bins the calling team’s signed streak; positive means the calling team itself has been on the run. Figure 2 makes clear just how heterogeneous the WP signal is across that dimension. Full timeouts called *into* a 6–9-point opposing run cost the caller -10.3 pp of WP ($G = 12,992$, $n = 15,219$); full timeouts called while the calling team is on its own 6–9-point run *add* $+9.5$ pp ($G = 1,135$, $n = 1,470$). The signed effects are roughly symmetric around streak = 0, and that symmetry is the headline finding here: timeouts do not reverse momentum within the post window. They ride it – regardless of which team called them.

Applying the same stratification to Q1 (Figure 3) gives a much weaker version of the same picture, because the PPP-gap baseline absorbs most of the signal. The direction still tracks: down-streak timeouts produce small positive Q1 excesses, up-streak timeouts produce small negative ones.

4.3 Stratified by Score Margin

Slicing on the calling team’s score margin (Figure 4) recovers a clean regression-to-mean signal in Q1 that pooling erases. Every down-by bucket for full timeouts shows positive Q1 excess (down15+: $+0.17$, $G = 24.4$; down6–15: $+0.11$, $G = 24.9$); every up-by bucket shows negative Q1 excess (up15+: -0.26 , $G = 39.5$; up6–15: -0.16 , $G = 37.8$). The same monotone pattern persists, more faintly, for inferred TV timeouts. The challenge subtype is too small to register a margin signal at all.

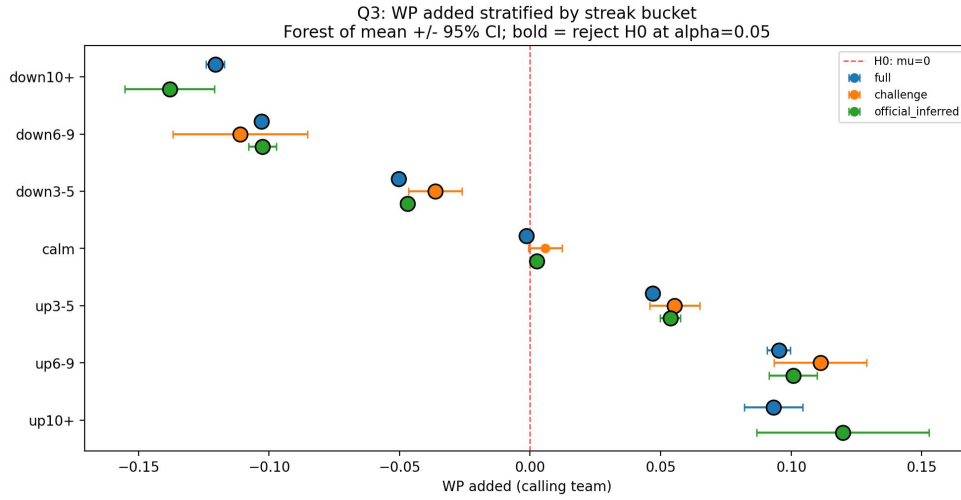


Figure 2: Q3 WP added stratified by pre-timeout streak. Bold filled markers indicate strata where $H_0: \mu = 0$ is rejected at $\alpha = 0.05$.

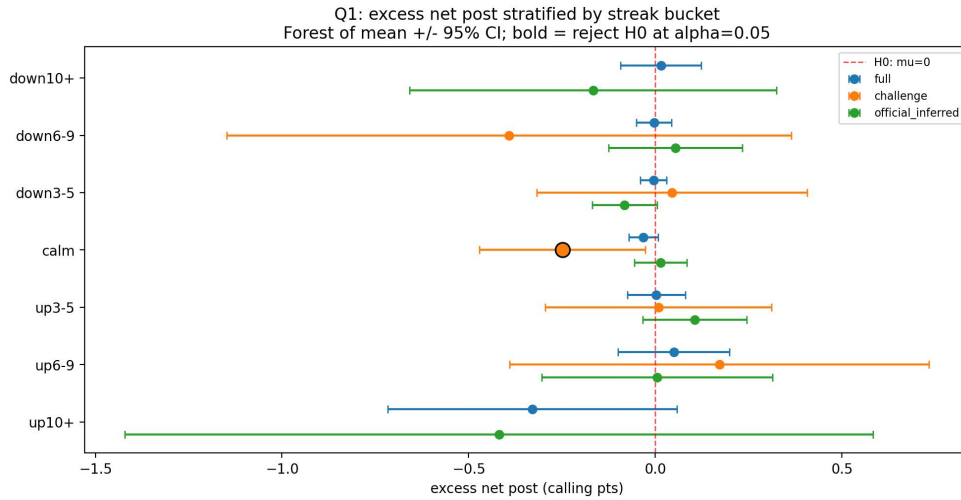


Figure 3: Q1 excess net post stratified by pre-timeout streak. Bold filled markers indicate strata where H_0 is rejected at $\alpha = 0.05$.

The corresponding Q3 margin stratification (Figure 5) tells a related story from the WP side. Full-timeout WP loss peaks in moderately adverse situations (**down6-15**: -6.1 pp, $G = 6,749$; **down1-5**: -5.3 pp, $G = 2,562$) and shrinks toward zero in extreme leads or deficits, where the outcome of the game is already mostly decided.

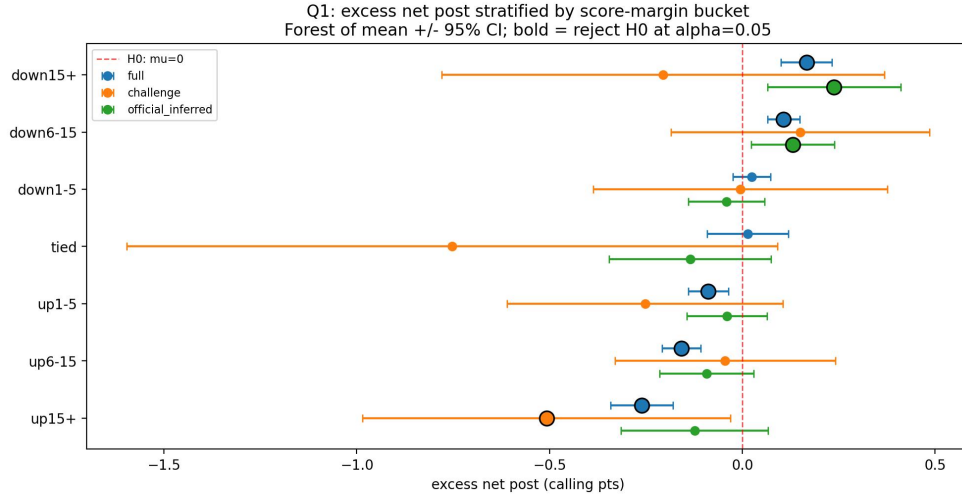


Figure 4: Q1 excess net post stratified by calling-team score margin. The left-to-right monotone trend is the regression-to-mean signature: trailing teams gain back roughly what they were missing, leading teams give some back.

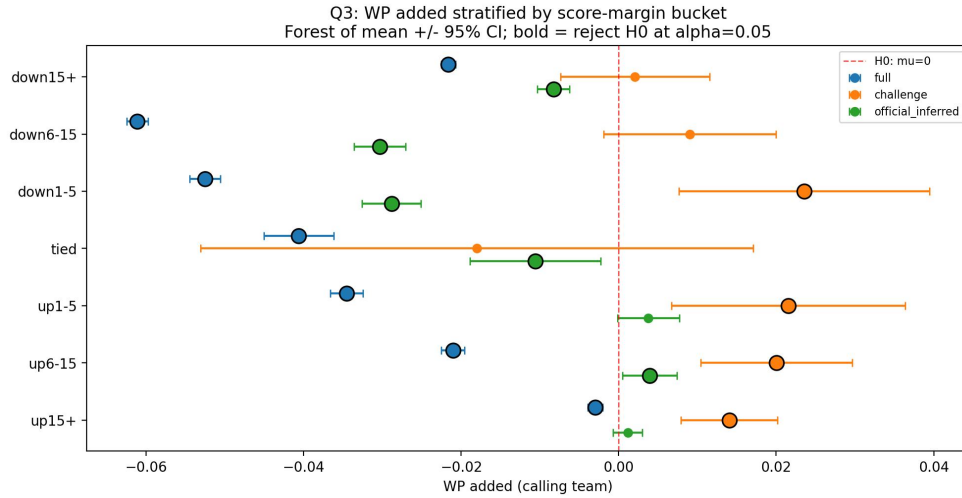


Figure 5: Q3 WP added stratified by calling-team score margin.

4.4 Stratified by Game Phase

Figure 6 breaks Q3 out by quarter, with Q4 split into early (> 5 min remaining) and clutch (≤ 5 min). The full-timeout WP cost is roughly flat across Q1–Q3 (-4.0 to -4.5 pp) and OT (-5.3 pp), but *halves* in Q4-clutch, to -2.0 pp. Two forces probably contribute to that drop: coaches are most deliberate about timeout spending in clutch time, and the WP model itself is most informative in those windows because each remaining possession carries

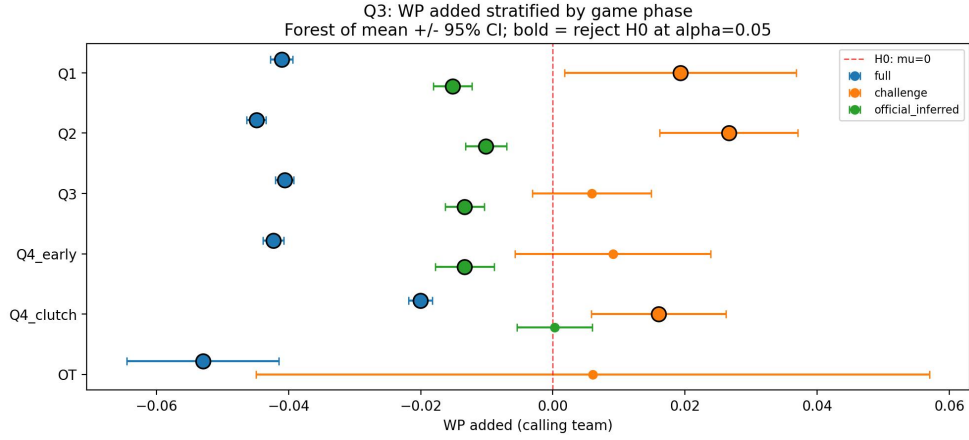
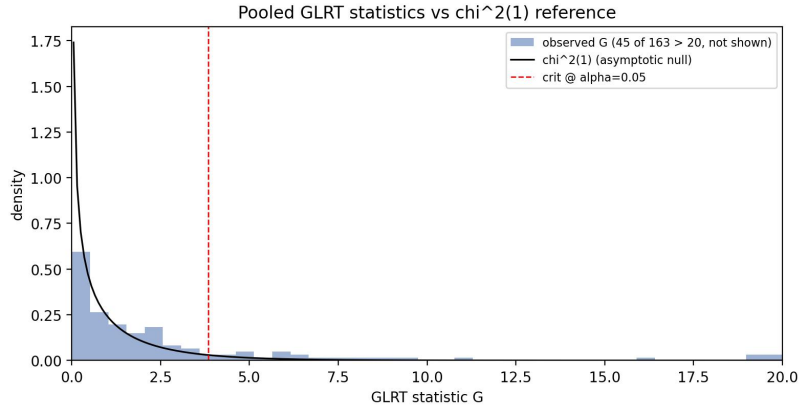


Figure 6: Q3 WP added stratified by game phase.


 Figure 7: Pooled empirical distribution of GLRT statistics across all 163 stratified tests, overlaid on the asymptotic χ^2_1 null reference.

more leverage. TV timeouts behave similarly – their WP cost collapses to zero in Q4-clutch (+0.0002 pp, ns).

4.5 Pooled Distribution of Test Statistics

Across four stratification schemes (overall, streak, margin, time) and three impact metrics we ran 163 stratified GLRTs in total, and 69 of them reject H_0 at $\alpha = 0.05$ – 42.3%, about $8\times$ the rate we would expect under a universal null. Figure 7 plots the empirical distribution of G values against its χ^2_1 reference. The shift to the right is heavy, which is what you would see if many of those rejections are genuine signal rather than chance.

5 Discussion

The three metrics behave so differently because each one packs the noise into the null in a different way. Q1 cancels by construction in the aggregate: the PPP-gap baseline is engineered so that $\mathbb{E}[\bar{X}] = 0$ under no-effect. Signal only re-emerges once we stratify on a context that the baseline does not already know about. Q2 leaves the opponent’s contribution in, which is why even the aggregate test rejects weakly. Q3 is fundamentally different: WP is non-linear in score margin, especially in the endgame, so a single ~ 1 -point swing can move WP by an order of magnitude more in clutch time than at tip-off. That is why even tiny mean shifts in WP produce GLRT statistics in the thousands.

The single clearest finding in the whole report is the signed-WP picture in Figure 2: *time-outs ride momentum, they do not reverse it*. Whether that reflects a genuine causal failure of the timeout (the calling team was hot, and would have stayed hot anyway) or selection bias (coaches call timeouts in specific, predictable game states) is something between-group GLRTs cannot answer on their own. What our tests do is pin down the *observed* pattern with high precision. Recovering the *cause* of that pattern requires matching, mediator analysis, or an instrument in the spirit of Weimer et al. (2023).

Limitations. A few worth flagging. First, the GLRT relies on the underlying metric being Gaussian. Q1 and Q2 are noisy averages of point counts and are approximately Gaussian by CLT; Q3 is the output of a logistic model and is bounded in $[-1, 1]$, so the Gaussian assumption breaks near the endpoints. In practice the t -test and GLRT p -values agree to four decimal places throughout, which is reassuring, but a non-parametric permutation test would be a more rigorous check. Second, the 163 tests are not independent – many events appear in multiple stratifications – so a Bonferroni or Benjamini–Hochberg correction would over-correct here; we report uncorrected p -values. Third, the TV-timeout cohort is rulebook-inferred rather than ground-truth labeled, with $F1 \approx 0.9$ against the validation era, so the `official_inferred` effect sizes should be read as a lower bound on the true mandatory-timeout effect.

Future work. The cleanest extension is to swap the unconditional null in Equation 1 for a counterfactual null that conditions on game state. The recipe: for each timeout, find a matched non-timeout possession in the same game, take the paired difference of the impact metric, and run the GLRT on those differences. That turns the GLRT into a within-pair causal estimator and keeps both the variance reduction of matching and the asymptotic distribution theory of the test.

Github Repository

The code for this project is available at <https://github.com/akseldkw07/NBA-Timeout-Impact>

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